Unit 1 Practice Test - Version 5

90 minutes | 60 marks | Circle one answer for Questions 1-10. Show reasoning for Questions 11-18. Label all graphs. Calculator permitted; formal differentiation rules are not required.

Part A - Multiple choice (10 marks)

1. Which relation is a function but is not one-to-one? [1]

A. {(0,1),(1,1),(2,3)} B. {(0,1),(0,2),(2,3)}
C. {(0,1),(1,2),(2,3)} D. {(0,0),(1,1),(2,2)}

2. What is the domain of f(x)=sqrt(2x-6)/(x-5)? [1]

A. [3,infinity) B. (3,5) U (5,infinity)
C. [3,5) U (5,infinity) D. (-infinity,3]

3. What is the range of g(x)=-|x+2|+5? [1]

A. [5,infinity) B. (-infinity,5]
C. (-infinity,5) D. all real numbers

4. If (9,3) lies on y=f(x), which point lies on h(x)=-3f(-2(x-4))+2? [1]

A. (-1/2,-7) B. (17/2,-7)
C. (-1/2,11) D. (4,-7)

5. For f(x)=(x-1)^2+2 restricted to x>=1, what is f^(-1)(x)? [1]

A. 1-sqrt(x-2) B. sqrt(x-1)+2
C. 1+sqrt(x+2) D. 1+sqrt(x-2)

6. Which function is odd? [1]

A. x^4-4 B. x^3-4x
C. x^3+4 D. x^2+x

7. Let p(x)=3x-1 for x<2 and p(x)=kx+1 for x>=2. For which value of k is p continuous at x=2? [1]

A. 1 B. 3/2
C. 2 D. 3

8. What is the average rate of change of f(x)=x^2-4x on [1,3]? [1]

A. 0 B. 2
C. -2 D. 4

9. The centred estimate of the instantaneous rate of h(t)=-t^2+6t at t=2, using [1.9,2.1], is: [1]

A. 0 B. 1
C. 4 D. 2

10. A secant from x=a to x=b has negative slope, while the tangent at x=a has slope 0. Which statement follows? [1]

A. The function is constant throughout [a,b]. B. The function has no values between a and b.
C. Its value at b is lower than at a, while it is flat at a. D. The tangent and secant must have equal slopes.

Optional rough work

Part B - Multipart questions (50 marks)

11. Piecewise functions [9 marks]

The function F is defined by

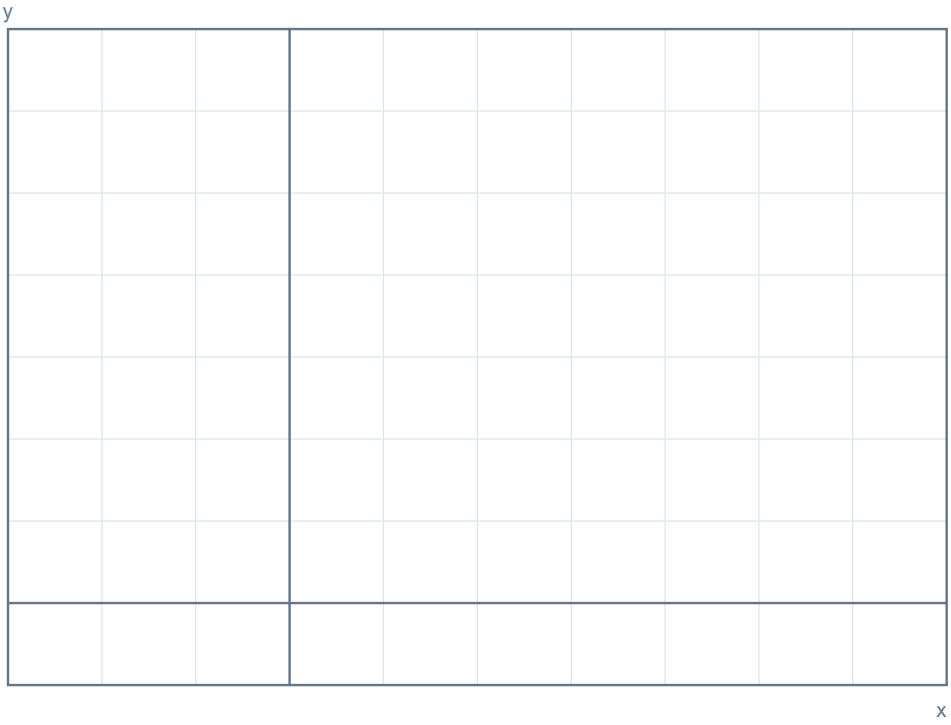
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-(x+1)^2+4 if -3 <= x < 0
F(x) = 2x+3 if 0 <= x <= 2
9-2x if 2 < x <= 5
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a. Sketch F on one set of axes. Label the vertex and all open and closed endpoints. [K2 C2]

b. State its domain and range in interval notation. [K2]

c. State the intervals where F is increasing. [K1]

d. Would the inverse relation be a function? Explain with one output that occurs for two different inputs. [T1 C1]



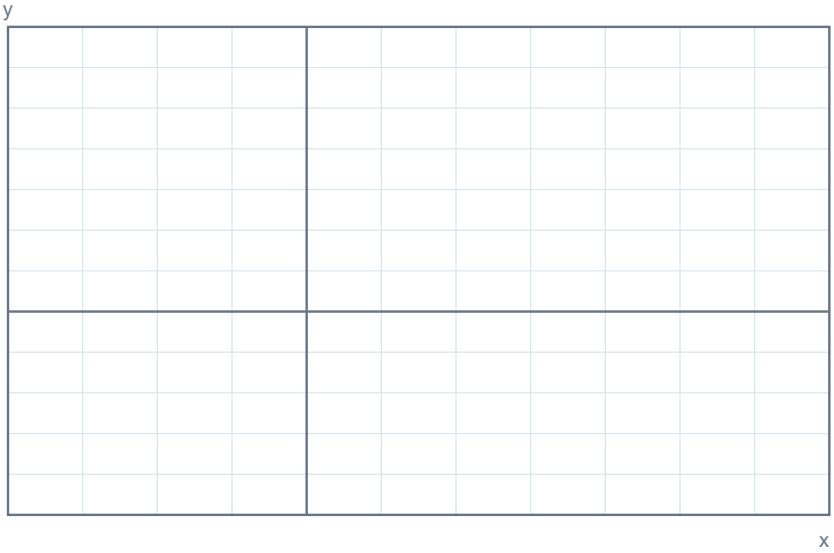
Domain: _____ Range: _____

Increasing intervals: _____

Inverse explanation:

12. Absolute value graph [6 marks]

Sketch g(x)=-3|x+1|+6. Label the vertex, both x-intercepts, and the y-intercept. [K2 C1] State the domain and range in interval notation. [K2] Explain how the coefficient -3 changes the graph of y=|x|. [T1]



Domain, range, and explanation:

13. Transforming a function [5 marks]

The point (9,3) lies on y=f(x), where f(x)=sqrt(x). Define h(x)=-3f(-2(x-4))+2.

a. Find the corresponding point on h, showing the input equation. [T2]

b. State the domain and range of h. [K2]

c. Describe the horizontal transformation. [C1]

14. Inverse relations [5 marks]

Let q(x)=-2(x-2)^2+7, initially with domain all real numbers.

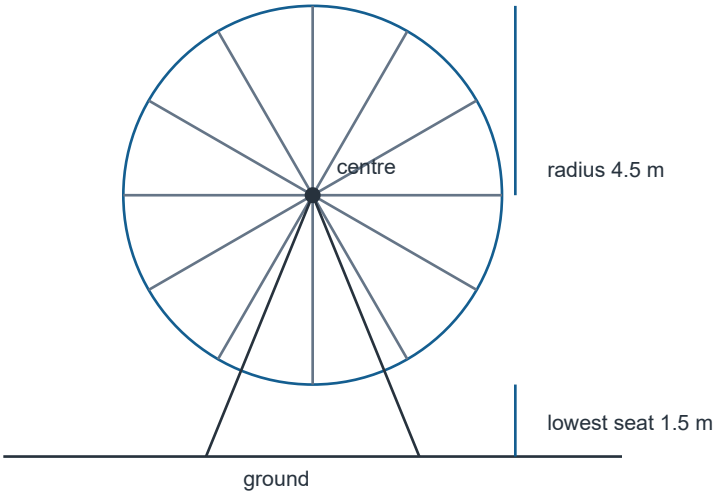
a. State the domain and range of its inverse relation. [K2]

b. Explain why this inverse relation is not a function. [T1]

c. Restrict the domain of q so that its inverse becomes a function, then write that inverse function and its domain. [A2]

15. Circular ride diagram [7 marks]

A circular ride has radius 4.5 m. Its lowest seat is 1.5 m above the ground. It makes exactly one rotation in 420 s at constant speed. A rider begins at the lowest point at $t=0$.

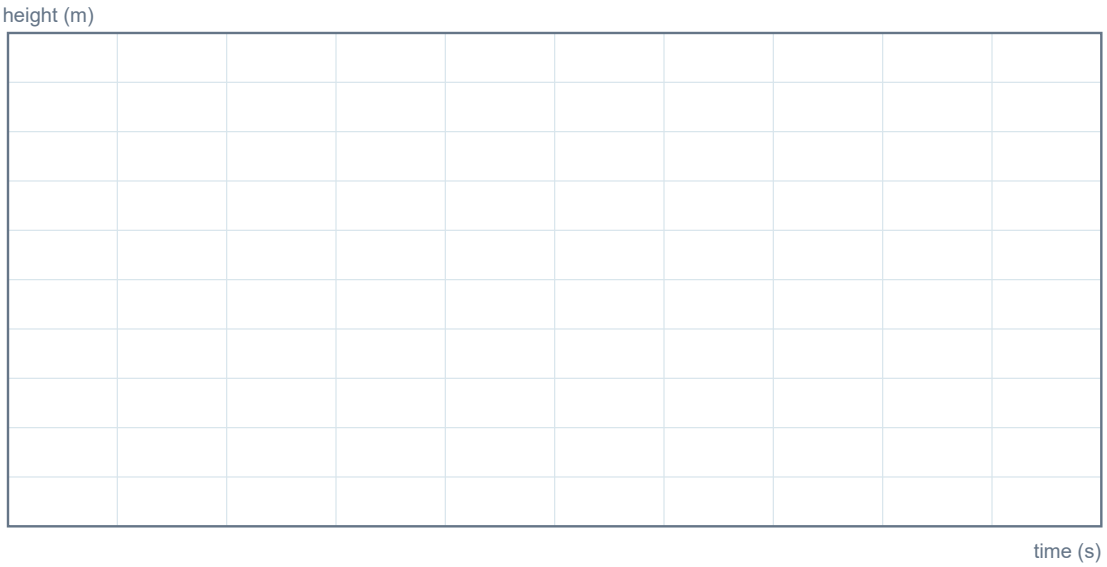


a. Determine the centre height and greatest seat height. [K2]

b. State the domain and range of the rider's height function for this ride, in interval notation. [K2]

c. A student says the greatest height is 6 m because $1.5+4.5=6$. Explain the error. [T1]

d. Sketch a qualitative height-versus-time graph for one rotation. Label the heights at $t=0$, 105, 210, 315, and 420 seconds. [C2]



Calculations and explanation:

16. Average and instantaneous rates [8 marks]

During its first 10 seconds, a ball's height above the ground is modelled by $H(t)=-t^2+10t+4$ metres.

a. Calculate the average rate of change of height over $[1,3]$. [A2]

b. Estimate the instantaneous rate at $t=3$ using the centred interval $[2.9,3.1]$. Show the two heights used. [A2 T1]

c. At what time does the ball reach its maximum modelled height, and what is that height? [K2]

d. Explain why your answers in a and b differ. [C1]

17. Graph properties and interval notation [5 marks]

Let $r(x)=x^4-3x^2+2$.

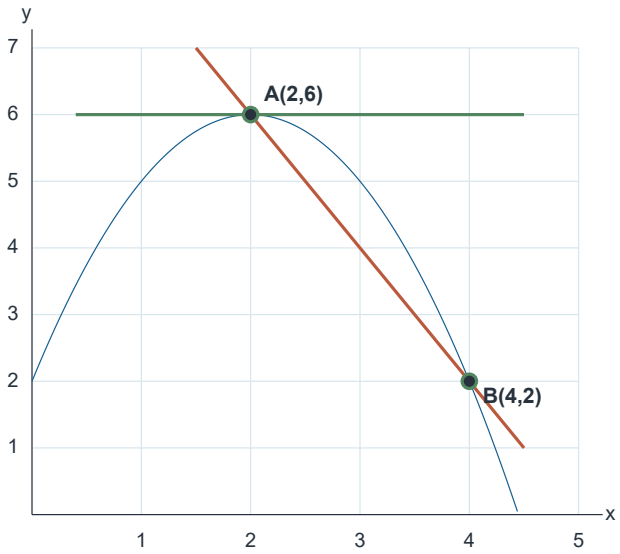
a. Show algebraically whether r is even, odd, or neither. [T2]

b. If $(2,6)$ lies on its graph, use symmetry to name one more point. [K1]

c. Write the solution to $|2x+1|<5$ in interval notation. [K2]

18. Secant and tangent diagram [5 marks]

The graph shows y=-(x-2)^2+6, a secant through A(2,6) and B(4,2), and the tangent at A.



a. Which line represents the average rate of change on [2,4]? Calculate its slope. [K1 A1]

b. Which line represents the instantaneous rate at x=2? State its slope. [K1 A1]

c. Explain why the average rate is negative while the instantaneous rate at A is zero. [T1]